COURSE STRUCTURE FOR B.TECH. Mechanical Engineering FIRST YEAR SEM – II (w. e. f. 2024-25)

SEMESTER-II (Subjects)				B.TECH. MECHANICAL ENGINEERING Sem-II											
Sr. No	Category Code	Course Code	Course Name	Teaching Scheme					Exam Scheme						
									Theory			Practical		Total	
				L	T	P	C	Hrs/wk	MS	ES	CE	LE	LE/Viva	Marks	
1	BSC	24MA103T	Mathematics - II	3	1	0	4	4	25	25	50	-	-	100	
2	BSC	24CH101T	Engineering Chemistry	3	0	0	3	3	25	25	50	-	-	100	
3	BSC	24CH101P	Engineering Chemistry Laboratory	0	0	2	1	2	25	25	50	-	-	100	
4	ESC	24CP102T	Computer Programming - II	1	0	0	1	1	-	-	-	25	25	50	
5	ESC	24CP102P	Computer Programming - II Laboratory	0	0	2	1	2	25	25	50	-	-	100	
6	ESC	24EE101T	Elements of Electrical and Electronics Engineering	3	0	0	3	3	-	-	-	25	25	50	
7	ESC	24EE101P	Elements of Electrical and Electronics Engineering Laboratory	0	0	2	1	2	-	-	-	25	25	50	
8	HSC	24HS103T	Indian Knowledge System	2	0	0	2	2							
9	HSC	24HS104T	Organizational Behaviour	1	0	0	1	1	25	25	50	-	-	100	
10	HSC	24HS105T	Professional Communication	2	0	0	2	2	25	25	50	-	-	100	
11	HSC	24YOG101	Yoga, Health and Hygiene	0	0	2	1	2	-	-	-	25	25	50	
		24NSS101	National Service Scheme (NSS)												
		24NCC101	National Cadet Corps (NCC)												
Total				**15**	**1**	**08**	**20**	**24**							

MS = Mid Semester ES = End Semester CE = Continuous Evaluation LW = Laboratory work LE = Laboratory Exam

Pandit Deendayal Energy University, School of Technology
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Semester – II

24MA103T					Mathematics – II					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs. / Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. Understand and apply principles of complex differentiation and integration effectively.
2. Solve systems, find eigenvalues, and apply transformations confidently.
3. Solve various ODEs, apply methods, and tackle engineering problems.
4. Analyze periodic functions, derive series, and apply in diverse applications.

UNIT 1 Complex Differentiation and Integration — 12 Hrs.

Limit, Continuity, Differentiability of the function of a complex variable, Analytic function, Cauchy-Riemann equation (in Cartesian coordinates), Harmonic function and its significance, Singularities, Definition of a Complex line integral, Contour integrals, Cauchy integral theorem, Cauchy Integral formula (CIF), CIF for derivatives, Taylor's series and Laurent Series , Calculation of residues, Cauchy Residue theorem, Applications of residues to evaluate real definite integrals..

UNIT 2 Matrix Algebra and Its Applications — 10 Hrs.

Solution of system of algebraic equation - Rank of a matrix, consistency of system of equation - Characteristic equation of a square matrix- Eigen values and Eigenvectors of a real matrix - Properties of Eigen values and Eigen vectors - Cayley-Hamilton theorem (without proof) - finding inverse of a matrix - Diagonalization of a matrix.

UNIT 3 Ordinary Differential Equations With Applications — 10 Hrs.

Differential equations of Higher-order differential equations with constant coefficients, Rules for finding C.F. and P.I., Method of variation of parameters, Cauchy and Legendre's linear equations, Simultaneous linear equations with constant coefficients (Operator method), applications of higher-order differential equations in solving engineering problems.

UNIT 4 FOURIER SERIES — 10 Hrs.

Periodic functions, Odd and even functions, Euler's formulae for Fourier series in an interval of length 2 pi, Change of interval, Dirichlet's conditions, Half range Sine and Cosine series, Complex Fourier series, Parseval's identity and its applications.

TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES:

At the end of the course, students will be able to:

CO1 : Recall fundamental concepts of complex differentiation, matrix algebra, differential equations, and Fourier series.
CO2 : Understand the significance of complex differentiation, integration, Matrix, ODE and Fourier series in respective contexts.
CO3 : Apply the concept of complex function, Matrix, ODE and Fourier series to extract the solutions of engineering problems
CO4 : Analyze the use of complex variable Matrix, ODE and Fourier series in engineering problems.
CO5 : Assess the significance and effectiveness of mathematical concepts and theorems in solving real-world problems, particularly in engineering and scientific applications.
CO6 : Design and construct solutions to complex mathematical problems using a variety of techniques, including transformations, mappings, and advanced methods in differential equations.

TEXT/REFERENCE BOOKS

1. R.V. Churchill and J. W. Brown, "Complex variables and applications", McGraw-Hill.
2. J. M. Howie, "Complex analysis", Springer-Verlag.
3. R. K. Jain and S. R. K. Iyengar, "Advanced Engineering Mathematics", Alpha Science.
4. Erwin Kreyszig, "Advanced Engineering Mathematics", John Wiley.
5. G. Strang, "Linear Algebra and its applications", Cengage Learning.
6. K. Hoffman and R. A. Kunze, "Linear Algebra", Prentice Hall of India.
7. B.V. Ramana, "Higher Engineering Mathematics", McGraw Hill.

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24CH101T					Engineering Chemistry					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	T			L		Total marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To develop the fundamental understanding about traditional materials.
2. To provide the knowledge about structural features, synthesis, properties of various categories of advanced materials.
3. To develop the knowledge and skills for different characterization techniques of materials.
4. To provide the knowledge about the role of chemistry in modern engineering applications.

UNIT I: CHEMISTRY OF ENGINEERING MATERIALS **12 Hrs.**

Traditional Materials: Introduction and classification of materials; metallic materials, polymeric, ceramic materials Advanced Materials: Introduction to nanomaterials: Properties and application; Carbonaceous materials (fullerene, carbon nanotube, graphene, etc.); Composite materials; Liquid crystals: Classification and Application

UNIT II: MODERN ANALYTICAL TECHNIQUES **10 Hrs.**

Instrumentation, principle and characterization of materials: X-ray diffraction (XRD), Electro analytical techniques; FTIR, UV-visible spectroscopy; Thermal analysis (TGA-DTA-DSC); Chromatographic techniques (GC, HPLC)

UNIT III: ADSORPTION, CATALYSIS AND KINETICS **10 Hrs.**

Adsorption - Characteristics, Classification, Application , Adsorption isotherms- Freundlich, Langmuir & BET Chemical Kinetics - Rate law, Arrhenius equation, Transition state theory, Collision theory; Complex reactions Catalysis - Homogeneous and Heterogeneous Catalysis; Mechanism of Catalysis; Industrial Applications of catalysts

UNIT IV: CHEMISTRY OF ENERGY DEVICES **10 Hrs.**

Principles of primary and secondary batteries, Fuel Cells and their operation principles, Principles and uses of supercapacitors; Photocatalytic hydrogen production: Principles and challenges; Traditional and new generation solar cells.

TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES:

At the end of the course, students will be able to:

CO1 : Remember the structural features and properties of different classes of traditional materials
CO2 : Classify advanced materials like nanomaterials, carbonaceous and composite materials, and liquid crystals
CO3 : Apply the skills by understanding various instrumental techniques for characterisation of materials.
CO4 : Analyse the key concepts in engineering chemistry viz. adsorption and chemical kinetics and laterally ponder over the applications of such concepts in engineering challenges.
CO5 : Justify the important insights into the industrial application of different types of catalysis via analysing mechanisms of catalysis.
CO6 : Develop the knowledge on the role of chemistry in various modern engineering applications such as in energy devices.

TEXT/REFERENCE BOOKS

1. W.D. Callister, An Introduction to Materials Science & Engineering, John Wiley & Sons.
2. MW Barsoum, Fundamental of Ceramics, IOP publishing.
3. T. Pradeep, Text book of Nanoscience and Nanotechnology, Mc. Graw Hill Education.
4. Murty, Shankar, B Raj, Rath, Murday, Textbook of Nanoscience and Nanotechnology", Springer.
5. V. Raghavan, Materials Science and Engineering, Prentice-Hall of India Private Limited.
6. A. Douglas, Donald Skoog, M.West, Principles of Instrumental Analysis, Cengage.
7. Jain & Jain, Engineering Chemistry, Dhanpat Rai Publishing Company.

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24CH101P					Engineering Chemistry Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To enhance and develop scientific and analytical skills
2. To relate concepts learned in chemistry and engineering to the real-world situations.
3. To acquire skills to perform laboratory experiments.
4. To demonstrate safe and proper use of standard chemistry glassware and equipment.

LIST OF EXPERIMENTS

1 **Iodometry**– To determine the strength of given copper sulphate solution by titrating against N/20 sodium thiosulphate (hypo) solution

2 **Iodimetry**– To determine the strength of given ascorbic acid by titrating against standard N/10 iodine solution

3 **Complexometric Titration**– To determine the total, permanent and temporary hardness of given water by complexometric titration using standard 0.01M EDTA solution

4 **pH metric titration**– To determine the strength of given HCl solution using a standard NaOH solution by performing a pH-metric titration

5 **Conductometric titration**– To determine the strength of given HCl solution using a standard NaOH solution by performing a conductometric titration

6 **Chemical Kinetics**– To study the kinetics of decomposition of sodium thiosulphate by a mineral acid

7 **Drawing chemical structures** - To Draw Chemical Structures of organic molecules using ChemDraw

8 **Colorimetric determination**: To determine the concentration of copper present in the effluent of electroplating industries by using colorimeter.

9 **Detection of biomolecule**: Detection of the presence of carbohydrates in test solution by using Benedict's reagent

10 **Preparation of drug molecule**: Preparation of Aspirin from salicylic acid

11 **Polymerization**– To prepare a polymer (Nylon 6,10), identify the functional groups by FT-IR

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Recall the concepts learned in chemistry and engineering to the real-world situations.
CO2 : Show the ability to identify, analyse and interpret the results from the experiments.
CO3 : Experiment with the instrumental method using conductometer and pH meter.
CO4 : Analyse compounds by titrimetric, gravimetric and instrumental methods.
CO5 : Determine the concentration of unknown solutions by spectrophotometric method.
CO6 : Predict the reaction rate and predict the order and rate constant.

TEXT/REFERENCE BOOKS

1. V. K. Ahluwalia, S Dhingra, A. Gulati, College Practical Chemistry, Universities Press.
2. J.B. Baruah, P Gogoi, Foundations of Experimental Chemistry, PharmaMed Press.
3. S. S. Sawhney, M. S. Jassal, S.P. Mittal, A Text Book of Chemistry Practical, Vol. I & II, APH Publishing Corp.

Pandit Deendayal Energy University, School of Technology
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Semester – II

24CP102T					Computer Programming II (For ME/CL/CH/BT)					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	0	1	1	25	50	25	-	-	100

COURSE OBJECTIVES

1. To develop a greater understanding of the foundational principles of the Python programming language
2. To acquaint with the data structures offered in Python.
3. To inculcate functional and logical problem-solving skills through programming.

UNIT 1: Fundamentals of Python Programming — 4 Hrs.

Introduction- History of python, Silent features, Advantage and Applications, Tools and its Framework, Variables, Input and Output

UNIT 2: Decision and Control Statements — 4 Hrs.

Operators in Python, Decision statements- If statements, If-else statements, Nested If-else statements, Control Statements - For, While, For-Else, While-Else, Continue and Break statements

UNIT 3: Data Structures and Collection — 3 Hrs.

String manipulation, List and slicing, Tuple, Set and Dictionary, Numpy and Dataframes

UNIT 4: Function and File Handling — 3 Hrs.

Built-in and user defined functions, Actual and Formal Parameters, Scope, Create, Read and Write text files

14 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the basics of python programming such as syntax, datatypes, variable declarations.
CO2 - Understand the fundamental of data structures in python such as list, tuple, set, dictionary and strings.
CO3 - Understand the use of built-in functions, custom modules and packages.
CO4 - Apply Python's features to solve problems and implement decision-making and loops.
CO5 - Evaluate the effectiveness of functions and judge Python code quality.
CO6 - Evaluate Python scripts, create functions, and design programs for file handling tasks.

TEXT/REFERENCE BOOKS

1. John V Guttag, Introduction to Computation and Programming Using Python, PHI.
2. Allen Downey, Jeffrey Elkner and Chris Meyers, How to think like a Computer Scientist, Learning with Python, Green Tea Press.
3. Martin C. Brown, Python: The Complete Reference, Osborne, McGraw-Hill.
4. R. Nageswara Rao, Core Python Programming", Dreamtech Press.

Pandit Deendayal Energy University, School of Technology
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Semester – II

24CP102P					Computer Programming II Laboratory (For ME/CL/CH/BT)					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	0	1	1	25	50	25	-	-	100

COURSE OBJECTIVES

1. To develop a greater understanding of the foundational principles of the Python programming language
2. To acquaint with the data structures offered in Python.
3. To inculcate functional and logical problem-solving skills through programming.

LIST OF EXPERIMENT

1. **Installation:** Installing Python on Windows, Mac, and Linux.
2. **Introduction of Different Tools & Framework:** Introduce and utilizes different tools and frameworks for python program development
3. **Fundamentals of Python Programming**
   Variable declaration, initialization, different syntax and keywords, Input/output operations
4. **Decision Statements**: Operators- Arithmetic, Logical, etc., Decision-making - if, if-else, and nested if-else statements.
5. **Control Statements:** Control statements such as for loops, while loops, and their application, Continue and break statement
6. **Data Structure:** Demonstrate string manipulation techniques, including string concatenation, list, slicing, and formatting.
7. **Python Collection:** Implement programs that illustrate the characteristics and usage of tuples, sets, and dictionaries, numpy, dataframes.
8. **Function:** Develop Python functions to perform common tasks, including mathematical operations, string manipulation, and data processing, Built-in and user-defined functions, parameter actual and formal parameters, Implement functions with different scopes.
9. **File Handling:** Develop Python scripts that create, read, and write text files, showcasing file handling operations such as opening, reading, writing, and closing files.

COURSE OUTCOMES

On completion of the course, student will be able to
CO1 - Apply Python basics tools, laying the foundation for practical usage.
CO2 - Apply Python's features to manipulate data structures effectively for solving real-world problems.
CO3 - Apply decision-making and looping to solve practical challenges.
CO4 - Analyze Python code to select appropriate constructs and data structures for optimal performance.
CO5 - Evaluate function effectiveness in Python programs for code quality and efficiency.
CO6 - Develop Python scripts and programs with robust file handling, utilizing modular functions for enhanced organization and reusability.

TEXT/REFERENCE BOOKS

1. John V Guttag, Introduction to Computation and Programming Using Python, PHI.
2. Allen Downey, Jeffrey Elkner and Chris Meyers, How to think like a Computer Scientist, Learning with Python, Green Tea Press.
3. Martin C. Brown, Python: The Complete Reference, Osborne, McGraw-Hill.
4. R. Nageswara Rao, Core Python Programming", Dreamtech Press.

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Semester – II

24EE101T					Elements of Electrical and Electronics Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart knowledge on DC and AC circuits.
2. To learn construction, working principle and characteristics of transformer and induction machines.
3. To introduce students to various means for electrical safety, protection of electrical installations and Batteries
4. To understand rectification through p-n junction diode, applications of diode and Transistor Characteristics

UNIT I: BASICS OF ELECTRONICS **10 Hrs**

Semiconductor Diodes and Applications, Diode as a Switch, Diode as a Rectifier, Half Wave and Full Wave Rectifiers with and without Filters; Breakdown Mechanisms, Zener Diode – Operation and Applications; Opto-Electronic Devices–LEDs, Photo Diode and Applications; Silicon Controlled Rectifier (SCR) – Operation, Construction, Characteristics, Ratings, Applications.
Transistor Characteristics covering, Bipolar Junction Transistor (BJT) – Construction, Operation, Amplifying Action, Common Base, Common Emitter and Common Collector Configurations, Field Effect Transistor (FET) – Construction, Characteristics of Junction FET.

UNIT II: DC CIRCUITS **10 Hrs**

Electrical circuit elements (R, L and C), voltage and current sources, dependent and independent sources, Ohms Law, temperature co-efficient of resistance, Kirchhoff current and voltage laws, voltage and current divider circuit, Thevenin and Norton Theorems and their equivalents, maximum power transfer and superposition theorems, nodal and mesh analysis, star-delta transformation, Time domain analysis/natural response of first order RL and RC Circuit.

UNIT III: AC CIRCUITS **10 Hrs**

Generation of AC voltage, representation of sinusoidal waveforms, RMS values of different sinusoidal waveforms, Rectangular and Polar representation of phasor, Sinusoid representation in time and frequency domain, Analysis of single-phase ac series circuits consisting of R, L, C, RL, RC, RLC combinations, instantaneous power, average power and reactive power, complex power and power factor.AC parallel circuit and its solution in admittance form, resonance in AC series and parallel circuits, polyphase circuits, star and delta representation of polyphase circuits, power measurement in polyphase circuits.

UNIT IV: INTRODUCTION TO ELECTRICAL MACHINES AND ELECTRICAL INSTALLATION **12 Hrs**

Magnetic materials and its B-H characteristic, Faraday's Law of Electromagnetic Induction; **Single Phase Transformers:** Construction, working principle, types, EMF equation, ideal transformer, practical transformer, Operation of transformer on no-load and load, **Induction Machine:** construction, types of 3-phase induction motors, working principle, production of rotating magnetic field, operation, starting and running torques, Torque-slip characteristics, Power Stages in an induction motor.
Components of LT Switchgear: Switch Fuse Unit (SFU), MCB, ELCB, MCCB, Types of Wires and Cables, Earthing, Types of Batteries, Important Characteristics for Batteries. Elementary calculations for energy consumption, Electrical safety rules.

s TOTAL HOURS **42 Hrs**

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Demonstrate application of different diodes in circuits and Understand the Transistor characteristics
CO2 – Understand electrical circuits using network theorems.
CO3 – Compare the behaviour of R, L and C and their combinations in AC circuits.
CO4 – Analyze balanced polyphase systems in star and delta configurations
CO5 – Understand the construction, working and basic characteristics of transformer and induction machines
CO6 – Recognize the importance of protective devices, electrical safety measures and characteristics of Batteries

TEXT/REFERENCE BOOKS

1. J. Bird, **"Electrical Circuit Theory and Technology"**, Routledge, Tailor and Francis Group
2. D. P. Kothari and I. J. Nagrath, **"Basic Electrical Engineering"**, Tata McGraw Hill
3. B. L. Theraja, **"Electrical Technology",** Vol. 1, S. Chand Publication, New Delhi.
4. Surjit Singh, **"Electrical Estimating and Costing"**, Dhanpat Rai and Co.
5. Boylestad and Nashlesky, "**Electronic Devices and Circuit Theory**", PHI

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24EE101P					Elements of Electrical and Electronics Engineering – Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To gain practical knowledge on DC and AC circuits
2. To learn operation of electrical instruments and electrical machines
3. To introduce students to various means for electrical safety, protection of electrical installations
4. To understand characteristics and applications of semiconductor diodes and transistors.

LIST OF EXPERIMENTS

1. Introduction to elements of electrical engineering laboratory and to study different electrical measuring instruments.
2. To Verify Ohm's law with linear resistors and find power dissipation in resistor.
3. To validate Thevenin's and Norton theorem for DC circuits.
4. To validate Superposition and Maximum Power Transfer theorem for DC circuits.
5. To evaluate the AC R, L and R-L series circuit performance and to measure the active power, the reactive power and the apparent power connected with single phase AC supply.
6. To evaluate performance of AC R-L parallel circuit and to measure the active power, the Reactive power and the apparent power connected with single phase AC supply.
7. To perform a direct load test on a single-phase transformer and determine the efficiency and voltage regulation at different loads.
8. To measure three phase power using two wattmeter method.
9. To draw the time-current characteristics of MCB and study the function of ELCB.
10. To obtain the VI characteristic of silicon and germanium diodes.
11. To obtain reverse characteristics of Zener diode.
12. To study half wave, full wave and bridge rectifiers.
13. To determine the DC Characteristics of BJT in CE Configuration.
14. To study the types of batteries and their characteristics

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – To understand the basic operation of electrical equipment's & measuring instruments.
CO2 – To perform various network theorems for DC circuits.
CO3 – To understand the performance of AC circuit with R, L load.
CO4 – To evaluate star and delta configuration of polyphase system and measure power in polyphase system.
CO5 – To evaluate the performance of single phase transformer.
CO6 – To understand the working principle of semiconductor diodes, transistor characteristics and its applications.

TEXT/REFERENCE BOOKS

1. J. Bird, **"Electrical Circuit Theory and Technology"**, Routledge, Tailor and Francis Group
2. D. P. Kothari and I. J. Nagrath, **"Basic Electrical Engineering"**, Tata McGraw Hill
3. B. L. Theraja, **"Electrical Technology",** Vol. 1, S. Chand Publication, New Delhi.
4. Surjit Singh, **"Electrical Estimating and Costing"**, Dhanpat Rai and Co.
5. Boylestad and Nashlesky, "**Electronic Devices and Circuit Theory**", PHI

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24HS103T					Indian Knowledge System					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	-	-	100

COURSE OBJECTIVES

1. To connect with the tradition of IKS
2. The students will be able to define Philosophical foundations of IKS:
3. To understand Fundamentals of Art and Architecture, Traditional and Historical Town Planning & Ancient Indian Art & Architecture

UNIT I: INTRODUCTION TO IKS **07Hrs**

What is IKS? Historical and philosophical foundations
Knowledge Framework and classifications in IKS
Indian scheme of knowledge , The knowledge triangle
Framework for establishing valid knowledge
Deductive or inductive logic framework
Potential fallacies in the reasoning process

UNIT II: IKS AND ENGINEERING DISCIPLINES **07Hrs**

Engineering marvels of ancient India: Civil engineering: Urban planning, water management
Materials engineering: Sustainable materials, traditional construction techniques Material science and Metallurgy in IKS
Mechanical engineering: Robotics and automation in ancient India
The role of IKS in addressing climate change challenge
IKS principles for resource management and conservation

UNIT III: IKS AND MATHEMATICS **07Hrs**

IKS and Indian Mathematics, Algebra, Geometry, Trignometry, Binary Mathematics, Magic
IKS and Indian Astronomy, Pañcāṅga – The Indian calendar system
Astronomical Instruments -Yantra

UNIT IV: IKS AND OTHER DISCIPLINES **07Hrs**

Indian Administration , Concept of state, Anushashan Parwa of Mahabharat,
Kautilya's Arthshastra
Social and Political Philosophies of Ancient India
IKS and Medicine: Ayurveda - A Holistic Approach to Health and Wellbeing

TOTAL HOURS: 28 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Define the fundamental concepts of Indian Knowledge System
CO2 – Describe various cost concepts of IKS relevant from engineering perspective.
CO3- To Apply Mathematics in the Vedas and Śulva SūtraSystems employed for representing numbers Spherical trigonometry &Celestial Sphere
CO4- To Analyse the science of Astronomy and the different units of time discussed in the texts
CO5- To be able to weigh the IKS systems vis s vis the current applications in various spheres of understanding
CO6To design applications and processes by incorporating thebl traditional knowledge to the present day concerns

TEXT/REFERENCE BOOKS

1. Introduction To Indian Knowledge System: Concepts and Applications by by B. Mahadevan, Nagendra Pavana, Vinayak Rajat Bhat.
2. The Vedas by Sri Chandrasekharendra Saraswati/Sankaracharya of Kanchi Kamakoti Peetham (Author)
3. Indian Knowledge Systems – Vol 1 & 2 by Avadhesh K. Singh, Kapil Kapoor